Distributed temperature measurements on the Juneau Icefield

Modeling temperature in terrestrial environments on the Juneau Icefield, either in periglacial areas such as nunataks and treelines or on glacier surfaces) could improve understanding of ecological and glacial processes by providing spatially distributed, rather than point, estimates of temperature fields. However, the material and capacity costs of deploying and maintaining high quality weather stations limits the availability of weather and climate observations to develop observed temperature fields (rather than point measurements) and also limits the verification of simulated fields. Over the last 15 years, the availability of relatively inexpensive temperature sensors (iButtons, LogTags, Hobo Pendants, etc.) has decreased the cost of recording temperatures at much higher spatial density than previously possible. However, the accuracy of these sensors depends on the environment in which they are deployed – properly shielding them from direct and reflected light (visible spectrum) and re-emitted infrared is critical to accurate temperature measurement. The bias associated with deployment conditions has been studied in some regions (tropical, see Terando et al. 2018; temperate, see Lundquist and Huggett 2008; mountain, see Navarro-Serrano et al. 2019) but to my knowledge never at high latitudes and never on glaciers. The potential to use such measurements depends on the magnitude and nature of this bias, and whether it can be mitigated sufficiently by appropriate shielding and deployment to be useful. Onset micro station data loggers with temperature probe(s) and shielding cost on the order of $500 a location, depending on sensors, with accuracy of <0.2C and 512k memory (sufficient to store 4 channels of observations for as long or longer than typical battery life of 2-4yr). LogTags ($25-$50, depending on volume order), Onset Hobot Pendants (~$65), and iButtons ($50-$65) are an order of magnitude less expensive, with tradeoffs in accuracy (typically <0.5C), storage volume (64k or less), and battery life (1-2yr).

Test deployments at Camp 17, Juneau Icefield Research Program

To attempt to evaluate bias measurements, we co-deployed three types of temperature sensors on temporary masts at two locations at Camp 17 (C17). Shielded (custom shields, see below) LogTag Trix16 temperature sensors were placed on masts collocated with shielded Onset temperature/relative humidity sensors (S-THC-M002 with Onset RS-3B radiation shields) and unshielded Onset temperature probes (S-TMB-M006). The Onset temp/rh sensor in shielded housing was secured with a hose clamp and white zip ties at the top of a ~2m mast, and the shielded LogTag was secured with steel wire to the mast just below the Onset temp/rh sensor, with the shield secured to the mast with white zip ties. The unshielded Onset probe was also secured to the mast with white zipties and serves as an unshielded reference. Unshielded surface temperature probes (Onset S-TMB-M006) were also deployed. I programmed all these sensors to record every 15 minutes, and they were launched on the hour, so should be coherent observations unless something is amiss. Time stamps were synced with computer clocks in mid-June, so there should be essentially no drift in observation time vs. actual clock.

The first mast collocated with the main C17 station on a rock fin at Camp 17 at approximately 1260m elevation (https://www.weather.gov/wrh/timeseries?site=LGCA2). The purpose of collocating with this station is to compare both the shielded Onset temp/rh and the shielded LogTag measurements to a higher standard temperature sensor. This mast is a ~2m, ~4cm diameter steel pipe secured in a rock pile. The second mast was deployed on the surface of Lemon Creek Glacier at an elevation of approximately 1130m, about 25 m SE of a mass balance stake. This mast is a ~3m, 2cm diameter PVC pipe with 1m set into the snow on the surface of the glacier and stabilized with a 0.7m x 0.5m insulation panel with snow piled on top and three polyrope guylines secured to the mast and anchored with 0.6m wood "deadmen" buried in the snow for stability, placed in a pattern with about 120 degrees of arc between anchors, with two guylines into prevailing wind and a third trailing. The Lemon Creek glacier mast has an additional unshielded LogTag wired to the mast for comparison purposes in direct and reflected sunlight common on sunny days on the glacier surface.

Custom shields for LogTags were developed for high(er) latitude regions because the seasonal light regimes in boreal and arctic regions challenge custom shielding developed for lower latitudes that rely on passive shielding from trees for successful measurement (e.g., Lundquist et al. 2008, Holden et al. 2011). Specifically, the trees used for shielding are smaller or non-existent, the light angles spring and fall are lower and shine into the "pagoda" shields (like those in Holden et al. 2011), and the increased hours of direct, overhead sunlight create a thermal environment where passive heating of the shielding, sensor, and surrounding landscape can bias temperatures higher than would be measured with more sophisticated (e.g., aspirated) instrumentation. In snow or glacier environments (or really any landscape with a high albedo), reflected sunlight from the surface also requires some form of blocking the sensor from below. The shields I've developed use a white plastic cannister covered with HVAC reflective tape, and inner, perforated cannister to promote airflow, a nylon washer to separate the tops of the nested cannister, and a white plastic funnel below the sensor to minimize reflected solar energy. Two pieces of steel wire hold the whole unit together and pull the LogTag up inside the housing; one goes through the hole in the LogTag and up through the inner perforated cannister, the nylon washer, and the outer cannister – if you leave it longer, you can then use the wire to secure it to the mast so the sensor doesn't slide down. The second wire attaches the upside down funnel – one end of the wire is bent up through two small holes in the funnel wall to secure it, then run parallel to the main wire up through the top of the cannisters and wrapped around the main wire above the outer cannister. These shields have compared reasonably well (~ near sensor accuracy on all but the warmest, calmest days and mean bias <1.0-1.5C) in boreal forest and treeline environments when compared to higher quality instruments. In "naturally aspirated" settings (windy, higher relative humidity) they do remarkably well, but it is yet unknown how they will compare over snow/ice.

Sensor downloads and materials

There is a yellow Right in the Rain field notebook in a Ziploc bag inside a small Rubbermaid action packer (black plastic box, gray lid) in the staff shack on a desk just to the left of the main entrance. That notebook has deployment times, serial numbers of sensors, and other notes. I (J. Littell) have a photo of this information should the notebook be lost. Any maintenance or problems that would result in erroneous / higher bias should be noted in the notebook, and the timing of removal and redeployment (see below) would be useful. If the serial numbers could be reassigned to the same experimental setting (e.g., glacier surface/reference station, shielded/unshielded, etc.) and noted in the notebook, that would be useful.

In the Action Packer are spare custom shields, LogTags, zipties, hose clamps, steel wire for replicating the deployment elsewhere. There are also all the cables and connectors needed to download the various sensors and communicate with them or reprogram them. Specifically:

- There is gray cable with an audio connector ("jack") on one end that has an RS232 (9pin) splice to USB (so two cables, already connected). You use this to connect the Onset 4-battery datalogger that has the shielded temp/rh probe and two unshielded temperature probes to a computer. On the bottom of the logger, there is a plastic port you unscrew with a flat bladed screwdriver and inside is the port the pin audio connector plugs in. The other end is USB and you connect that to a computer with the HoboWare software running. It should detect the logger and you can choose to read out the data series and export the file as a .csv somewhere convenient on the computer.
- There is a "cradle" with USB cable labeled "LogTag"; you use this to read out the LogTag data in the LogTag Analyzer software (on one Seth's computers).
- There is also a little box called an Onset U30 shuttle and its two cables – one about 15cm long with two audio pins, and another much longer with a micro USB on one end and a regular USB on the other end. The pin-to-pin one allows you take the box out in the field and plug it in to the logger (same port as above) and download the data. The longer one allows you to program the shuttle or read it out from a computer. There is a chance that the Hoboware Lite software installed on the computers cannot talk to the shuttle – you may need HoboWare Pro, in which case you'd be out of luck trying to download the data that way. You should be able to download the loggers directly using the combined cable above with a laptop with the Lite software though.
- Serial numbers are on stickers on the boxes of Hobo Onset Microstations (the logger number), and stamped on the cables for each sensor. Serial numbers for LogTags are on the back of the sensor.

Station maintenance

Sensors are self-tending and should not need any regularly scheduled maintenance for the duration of JIRP stay at C17. It is possible, however, given the temporary nature of the masts, that wind events or melt (in the case of the glacier surface station) could de-stabilize the deployment, and it would be useful to fix that should it occur so that observations are as robust as possible. The glacier station, in particular, may melt out rapidly and the mast base and "deadman" anchors could need attention every few days, perhaps more frequently.

At some point before departing for C10, the sensors should be detached from the masts and stashed in the Action Packer and flown to C10. You can use a pair of wire snips (in the Generator Shed) to carefully cut the zipties and steel wires (trash these, please, or put them in a bag in the Action Packer, but don't leave on the glacier or in C17) and carefully remove the cables, sensors, and housings from the masts. You do NOT need to unplug or dismantle the Onset Hobo Micrologger's various cables – just leave them running and put the box ad its cables (easier if you coil and zip tie them neatly but not too tightly) in a Ziploc and in your pack and then (unsealed Ziploc) in the action packer. It would probably be advantageous to bring the poly rope guylines and shock cord bungees to the next camp, though you will probably have to find mast materials and anchors at C10 as best you can. The masts go up in the back of the Gen Shed – you'll see a bundle of similar PVC there and that's also where I found the steel pole.

At C10, you should be able to replicate the deployment (one on the snow/ice surface, one near some other more permanent instrumentation). At that point, you'll want to figure out with Jessica Badgeley how to incorporate the iButtons she brings into this scheme. There's no reason you can't use the spare shields for iButtons, but she may have other ideas. In any case, you have the tools here for a medium quality comparison (the Hobo temp/rh probe) and hopefully high-quality data from the C17/C10 permanent instrumentation.

Data use

You are free to use the resulting datastreams to do whatever you want – you get the gist of what I was thinking above (essentially, bias comparison to determine the utility of measurements form these sensors) but there are lots of other possibilities. I'm a fan of academic inclusivity, so let's collaborate. Eventually, I'd like the LogTags, cables, and instrumentation back and plan to use the data as one among several other already complete bias estimates for a paper. I'd be happy to include any and all on such a publication if there is interest. Annie Boucher and I can correspond about the best way to get these, but a base strategy would be to fly them back out to Juneau after C10 or C18 so they can stay in the Eagle Valley Center until I can come get them this fall or next year.

Jeremy Littell
jlittell@usgs.gov
907-360-9416

4210 University Drive
Anchorage, AK 99508

References

Holden, Z. A.; Klene, A. E.; Keefe, R. F.; Moisen, G. G. 2013. Design and evaluation of an inexpensive radiation shield for monitoring surface air temperatures. Agricultural and Forest Meteorology. 180: 281-286. https://doi.org/10.1016/j.agrformet.2013.06.011

Lundquist, J. D., and Huggett, B. (2008), Evergreen trees as inexpensive radiation shields for temperature sensors, *Water Resour. Res.*, 44, W00D04, doi:10.1029/2008WR006979.

Navarro-Serrano F, Lopez-Moreno J, Azorin-Molina C, et al. (2019) Air temperature measurements using autonomous self-recording dataloggers in mountainous and snow-covered areas. Atmospheric Research 224: 168–179. https://doi.org/10.1016/j.atmosres.2019.03.034

Terando, A. J., Youngsteadt, E., Meineke, E. K., & Prado, S. G. (2017). Ad hoc instrumentation methods in ecological studies produce highly biased temperature measurements. *Ecology and evolution*, *7*(23), 9890–9904. https://doi.org/10.1002/ece3.3499